

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act
VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI
For

MASTERS IN GEOINFORMATICS

(Two Year Degree Programme, 1st & 2nd Semester System)

NEP 2020
Department of Geography, Panjab University Chandigarh
Implemented from the Academic Session 2026-27

Examinations, 2026-27

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APPLICABILITY OF REGULATIONS FOR THE TIME BEING IN FORCE

Notwithstanding the integrated nature of a programme spread over more than one academic year, the regulations in force at the time a student joins a course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all students whether old or new.

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ABOUT THE COURSE

Teaching and learning of geoinformatics at the Department of Geography,Panjab University Chandigarh has reached a milestone. The department has been pioneer in introducing professional programme on geospatial technology. We introduced PG Diploma in Remote Sensing and Geographic Information System in 2001 which was later upgraded to a two-year Masters in Remote Sensing and Geographic Information System. Keeping pace with technological advancements in the field, the programme was further restructured and renamed as Masters in Geoinformatics in 2018. This is a multi-disciplinary and professional programme being offered since the academic session 2019-20; the most recent upgradation in course content has been introduced w.e.f. academic session 2026-27. The programme is robust in terms of content and professional flavour as well as innovative, flexible and interdisciplinary in nature. This two-yearprogramme divided into four semesters aims at developing scientific and professional temperament among students.

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WHY STUDY WITH US

Department of Geographyoffers a postgraduate programme thataims at developing technically skilled and scientifically oriented professionals in the field of geoinformatics. The programme is dynamic, uniquely designed and flexible that not only produce highly trained professionals but also encourages innovation and out-of-box thinking.Our students get exposed to a wide range of learning and character-building practices. We offerskill enhancement, conceptual and theoretical clarity, technical capabilities, analytical and decision-making qualities.Through a focussed field-basedand applied approach, we prepare students to develop practical and operational understanding of spacetechnology.

The highlights of our curriculum are:

- Conceptual & Theoretical Clarity of Key Concepts in Geoinformatics.
- Exposure to Remote Sensing and Digital Image Processing.
- Technical skills in the field of Photogrammetry.
- Innovative learning of GIS andGNSS.
- Understanding of basic Computer Skills and Programming.
- Development of Analytical and Creative Skills in Spatial Analysis.
- Specialization in Geosciences, Urban & Regional Planning and Disaster Management.
- Upgradation of Skills for Field-work Based Data Collection of Geo-Information.
- Improvement in Quantitative and Qualitative Research Skills.
- Project Formulation and Management.
- Report Writing and Delivering High Quality Output.

ELIGIBILITY CRITERIA

- The eligibility for this course is Bachelor's Degree with Geography/Geology/Geophysics/ Mathematics/ Physics/Botany/Environment Science/Computer Science/Urban Planning/Regional Planning/B. Tech./B.C.A. or Master's Degree in Geography/Geology/ Geophysics/ Mathematics/Physics/ Botany/Environment Science/Computer Science/M.C.A./Urban

Planning/Regional Planning with at least 50% marks in aggregate. The admission to the course shall be through an Entrance Test.

- All candidates, except those sponsored by the Government/Semi-Government Organisations and Institutions, will be required to qualify the Entrance Test conducted by the Panjab University, Chandigarh. The apportioning of weightages for the purpose of preparing the merit list will be as follows:

  Entrance Test: 50%
  Qualifying Examination (Bachelor's Degree): 50%
- Students who have studied Geography at the B.A./B.Sc./M.A./M.Sc. level shall be given an additional weightage of 15 per cent of the academic score at Bachelor's level.
- Academic and other weightage(s), if any, shall be based on the percentage of marks obtained by the eligible candidates in the Qualifying Examination (Bachelor's Degree) as prescribed and admissible under Panjab University Rules.

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students:
   - Terminal Evaluation:80% (or as specified in the syllabus)
   - Continuous Assessment: 20 % (or as specified in the syllabus)
   - Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.
   - In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct the following:
     - Written Test: 25
     - Snap Test: 25
     - Participation in Class discussion: 15
     - Term Paper: 25
     - Attendance: 10

     Total: 100 (reduced to 20/10 or as specified in the syllabus)
2. Weightage of 2/1 mark(s) for attendance component out of 20/10 marks for continuous assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/workshops. The break-up of marks for **attendance component** for theory papers shall be as under:

Attendance Component	***Mark/s for Theory Papers***
(a) 75 % and above up to 85 %:	1 (if total assessment is of 20 Marks)
(b) Above 85 %:	2 (if total assessment is of 20 Marks)
3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20/10, will be carried forward and added to his/her score out of the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.
4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch

PROGRAMME OUTLINE

SEMESTER-I

Course Type	Course Code	Course Title	Max. Marks	Credit	Hours per Week
Core Course (Theory)	GI 101	Fundamentals of Remote Sensing	50	2	2
	GI 102	Geographic Information System	50	2	2
	GI 103	Basics of GNSS and Surveying	50	2	2
	GI 104	Fundamentals of Computer Programming	50	2	2
Core Course (Practical)	GI 105	Introduction to Image Processing	50	2	4
	GI 106	GIS Software and Database Management	50	2	4
	GI 107	GNSS and Total Station Survey	50	2	4
	GI 108	Open-Source GIS	50	2	4
	GI 109	JavaScript and SQL	50	2	4
	GI 110	Statistical Techniques and Analysis	50	2	4
		Semester Total	***500***	***20***	

SEMESTER-II

Course Type	Course Code	Title	Max. Marks	Credit	Hours per Week
Core Course (Theory)	GI 201	Introduction to Cartography	50	2	2
	GI 202	Spatial Statistics and Analysis	50	2	2
	GI 203	Advances in Digital Image Processing	50	2	2
	GI 204	Fundamentals of Photogrammetry	50	2	2
Core Course (Practical)	GI 205	Cartographic Techniques and Visualisation	50	2	4
	GI 206	Spatial Analysis Tools and Techniques	50	2	4
	GI 207	Digital Image Processing Techniques	50	2	4
	GI 208	Photogrammetry and Drone Technology	50	2	4
	GI 209	Programming in Python and R	50	2	4
	GI 210	Web GIS	50	2	4
		Semester Total	***500***	***20***	

SEMESTER-III

Course Type	Course Code	Title	Max. Marks	Credit	Hours per Week
Core Course (Theory)	GI 301	Geoinformatics for Geosciences	50	2	2
	GI 302	Geoinformatics for Disaster Studies	50	2	2
	GI 303	Geoinformatics for Urban and Regional Planning	50	2	2
RBC-RM/DST	GI 304	Research Methodology	50	2	2
	GI 305	Research Report Writing	50	2	2+2*
Core Course (Practical)	GI 306	Geomorphic, Hydrological and Environmental Modeling	50	2	4
	GI 307	Disaster Risk and Climate Change Modeling	50	2	4
	GI 308	Geoinformatics for Urban Studies	50	2	4
	GI 309	Internship and Internship Report (Training/Field Work)	100	4	2+4*
		Semester Total	***500***	***20***	

SEMESTER-IV

Course Type	Course Code	Title	Max. Marks	Credit	Hours per Week
RBC-DST	GI 401	Dissertation 1. Formulation of Research Proposal 2. Mid Term appraisal 3. Pre-submission Presentation 4. Dissertation Evaluation 5. Presentation 6. Viva-voce	 50 75 75 200 50 50	20	2+4*
		Semester Total	***500***	***20***	
		GRAND TOTAL (Semester-I, II, III, IV)	**2000**	**80**	

NOTE: * 2/4 hours per teacher supervising the Internship/Research Report/Dissertation

Note:

1. After successful completion of first two semesters, there shall be a provision for the candidate/s to opt out of the Course. In such an instance S/He will be awarded Diploma in Geoinformatics.
2. The minimum marks required to pass the examination shall be 50% in aggregate in each semester and 45% in each individual paper: (i) theory paper, (ii) practical examination, (iii) internship report (iv) research report and (v) dissertation.
3. A candidate, who fails in a particular semester, shall be allowed to re-appear in the subsequent semester with a maximum of two chances to be availed consecutively for the respective semester.
4. A candidate who fails in one or more theory and/or practical/internship report/research report/ dissertation examination/s shall be allowed to re-appear in the subsequent semester with a maximum of two chances to be availed consecutively in the respective theory and/or practical/internship report/research report/dissertation examination/s.
5. The Internship Report shall be prepared by the candidate on the basis of work done or experience gained at Remote Sensing Centre(s)/government/private institute(s)/companies/industry in India. The candidate shall submit three copies of his/her Internship Report ten days before the commencement of the theory examination of the said semester.
6. In case of Research Report/Dissertation, the Board of Control shall assign Supervisor/Co-supervisors for each candidate for supervising the candidate on an approved topic. External Supervisor/Co-supervisors may be allotted, subject to their eligibility to be determined by the Board of Control.

SEMESTER-I

GI 101
FUNDAMENTALS OF REMOTE SENSING

Course Code: GI 101	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUT COMES: The course would help students able to:

- Define and explain fundamental conceptsof remote sensing
- Evaluate and summarise satellite sensors and their applications
- Apply tools and techniques to extract and interpretinformation from satellite imageries

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

Fundamental Concepts of Remote Sensing

i. Remote Sensing: Concept and Types
ii. Remote Sensing Platforms, Sensors and Scanning Systems
iii. Applications of Remote Sensing, Major Satellite Systems

Unit-II

Energy Principles and Interaction Mechanisms

iv. EMR and Radiation Principles
v. Energy-Atmosphere Interaction; Atmospheric Windows
vi. Energy-Earth Interaction; Spectral Signatures of Surface Features

Unit-III

Image Processing and Interpretation

vii. Satellite Image: Meaning, Types and Characteristics
viii. Concept and types of Resolution
ix. Elements of Image Interpretation; Basics of Image Processing

Unit-IV

Advances in Remote Sensing

x. Thermal Remote Sensing: Concept, Mechanism and Application
xi. Microwave Remote Sensing: Concept, Mechanism and Application
xii. Hyperspectral Remote Sensing: Concept, Mechanism and Application

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing*, Guilford Press, 2012.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.

3. Joseph, G. *Fundamentals of Remote Sensing*, Second Edition, Universities Press, 2005.
4. Lillesand, Kiefer and Chipman, *Remote Sensing and Image Interpretation*, Wiley, 2015.
5. Sabins, Jr. F.F. and Ellis, J.M., *Remote Sensing: Principles, Interpretation, and Applications,* Fourth Edition. (2020), N.p.: Waveland Press.
6. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

Suggested Readings:

1. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
2. Dong, P. and Qi Chen, LiDAR Remote Sensing and Applications. CRC Press, 2018.
3. Mather, P.M. and M.G. Koch, *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
4. Rees, W.G., *Physical Principles of Remote Sensing,* 3rd Ed., Cambridge University Press, 2012
5. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.

GI 102
GEOGRAPHIC INFORMATION SYSTEM

Course Code: GI 102	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Describe key concepts of geographic information system
- Analyse components, elements and applications of GIS
- Develop ability to collect, format, store, and edit spatial data
- Apply GIS tools and techniques for geodatabase creation and management

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit- I

GIS Concepts and Definitions

i. GIS: Concept, Significance and Developments
ii. GIS Components: Hardware and Software
iii. Applications of Geographic Information Systems

Unit-II

Understanding Coordinates and Projections

iv. Coordinate Systems: Geographic and Projected Coordinates
v. Concept of Datum; Ellipsoid, Spheroid and Geoid Model
vi. Projections and Transformation

Unit-III

Spatial Data and Data Management

vii. Nature of Spatial data: Points, Lines and Polygons
viii. Spatial Data Models: Nature and Characteristics of Raster and Vector Data
ix. DBMS; Data Structure Models: Hierarchical, Network, Relational, Object Oriented

Unit-IV

Data Analysis and Information Extraction

x. Data Acquisition, Input, Editing, Topology Creation
xi. Data Query (Spatial & Non-Spatial) and Feature Extraction
xii. Proximity and Overlay Analysis

LIST OF READINGS

Essential Readings:

1. Longley, P. A., Goodchild, M. F., Maguire, D. J., Rhind, D. W. *Geographic Information Science and Systems*. Germany, Wiley, 2015.
2. Heywood, I., Sarah Cornelius and Steve Carver, *An Introduction to Geographical Information Systems*, 3rd Edition, Pearson, Dorling Kindersley (India) Pvt. Ltd, 2010.
3. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.
4. Burrough, P. A., McDonnell, R. A., Lloyd, C. D. *Principles of Geographical Information Systems.* United Kingdom: OUP Oxford. 2015
5. Maguire, D.J., M. Batty and Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.

Suggested Readings:

1. Bolstad, P.: *GIS Fundamentals: A First Text on Geographic Information Systems*, XanEdu Publishing Inc, 2016.
2. DeMeers, M.N., *Fundamentals of Geographic Information Systems*, John Wiley and sons, 2008.
3. ESRI: *Esri Map Book*, Volume 32, Esri Press, 2017.
4. Graser, A.: *Learning QGIS: Create great maps and perform geoprocessing tasks with ease*, 3rd. ed., Packt Publishing, 2016.
5. Wise, S., *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.

GI 103
BASICS OF GNSS AND SURVEYING

Course Code: GI 103	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Describe the components, characteristics and principles of GNSS
- Analyse the influence of various sources of error on positioning accuracy
- Develop skills to use GNSS and Surveying tools and techniques
- Execute GNSS and Total Station based surveys

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

i. GNSS: Concept, Characteristics, and Component
ii. Principles of Operation, Accuracy and Errors
iii. Applications global navigation satellite system

Unit-II

iv. Global Navigation Satellite Systems: GPS, GLONASS, GALILEO, Beidou
v. Regional Navigation Satellite Systems: IRNSS, QZSS
vi. Integration of various GNSS constellations

Unit-III

vii. Concept, Components and types of DGPS
viii. Surveying with GNSS: Precise Point Positioning (PPP) and Real-Time Kinematics (RTK)
ix. GNSS Augmentation Systems: SBAS and GBAS.

Unit-IV

x. Surveying: Methods and Classification
xi. Introduction to Total Station; Equipments and Components
xii. Measurements of Distance, Angle and Elevation

LIST OF READINGS

Essential Readings:

1. Christopher J. Hegarty, Elliott D. Kaplan (Eds.). *Understanding GPS/GNSS: Principles and Applications,* Third Edition. United Kingdom: Artech House Publishers.2017.
2. El-Rabbani, A. *Introduction to GPS: The Global Positioning System*, Artech House, 2002.
3. Gopi, S. *Advanced Surveying: Total Station, GIS and Remote Sensing.* India: Pearson Edu. 2007.
4. Hofmann-Wellenhof, B., Lichtenegger, H., Wasle, E. *GNSS–Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and More*. Austria: Springer Vienna. 2007
5. Leick, A., Rapoport, L., Tatarnikov, D. *GPS Satellite Surveying*. Germany: Wiley. 2015
6. Walker, J. and Awange, J. *Surveying for Civil and Mine Engineers: Acquire the Skills in Weeks*. Germany: Springer International Publishing. 2020.

Suggested Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009
2. Kavanagh, B. F.&Mastin, T. B. *Surveying: Principles and Applications*. UK: Pearson. 2014
3. Rao, G.S., *Global Navigation Satellite Systems- With Essentials of Satellite Communications*, Tata McGraw Hill Education Private Limited, New Delhi, 2010.
4. Schofield, W. and Breach, M. *Engineering Surveying*. United Kingdom: CRC Press. 2007
5. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

GI 104
FUNDAMENTALS OF COMPUTERPROGRAMMING

Course Code: GI 104	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Comprehend basic concepts of computer programming
- Relate computer programming applications in geoinformatics
- Analyse and apply programming tools and techniques in geoinformatics

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit-I

Computer Concepts OS and Application Software

i. Computer Systems: Hardware, Software, Input and Output Devices
ii. Data Representation; Fundamental Concepts of Operating Systems.
iii. Windows Operating System; MS-Word and MS-Excel Software.

Unit-II

Computer Graphics, Computer Networks and Web Application Development

iv. Fundamentals of Computer Graphics, Image Formats; Computer Networks
v. Internet and its services,Search Engines, Techniques to search effectively
vi. Introduction to web application development using HTML

Unit-III

Data Concepts and Information Systems

vii. Key Database Concepts; Database Management Systems, Type of DBMS
viii. Structured Query Language (SQL); Data Visualization, Data Mining
ix. Introduction to Information Systems; Types of Information Systems

Unit-IV

Programming Concepts

x. Concept of a Program, Data Types, Keywords, Structure of a Program.
xi. Writing 'C' Programs:Statements, Types of Statements
xii. Library Functions and User Defined Functions, Arrays, File Handling.

LIST OF READINGS

Essential Readings

1. Aitchison, A. *Beginning Spatial with SQL Server 2008.* United States: Apress. 2009
2. Basandra, S. K., *Computers Today*, Galgotia Publications Pvt. Ltd., 2010.
3. Frye, C., *Microsoft Excel 2013: Step by Step*, Microsoft Press, 2013.
4. Lambert, J. and J. Cox, *Microsoft Word 2013: Step by Step*, Eastern Economy Edition, 2013.
5. Miller, M., *Easy Computer Basics, Windows 8.1 Edition*, Pearson Education, Inc. 2014.
6. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
7. Zandbergen, P. A. *Python Scripting for ArcGIS.* United States: ESRI Press. 2013.

Suggested Readings

1. Freedman, J., *Microsoft Word 2013: Plain & Simple*, O'Reilly Media Inc., 2013.
2. Lovelace, R., Nowosad, J., Muenchow, J. *Geocomputation with R.* US: CRC Press. 2019.
3. Lutz, M. *Programming Python*, 3rd Edition, O'Reilly Media, Inc. 2006.
4. Norton, P., *Computing Fundamentals*, Student Edition, McGraw-Hill, 2004.
5. Tateosian, L. *Python for ArcGIS*, Springer, 2015
6. Yang, C. *Introduction to GIS Programming and Fundamentals with Python and ArcGIS*, CRC Press, 2017.

GI 105
INTRODUCTION TO IMAGE PROCESSING

Course Code: GI 105	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Learn the applications of image processing software
- Apply image processing techniques for analysing satellite imageries
- Interpret satellite imageries and extract information

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Measuring Satellite Orbital Characteristics
2. EMR: Wavelength, Frequency and Spectrum
3. Spectral Signature of Surface Features
4. Visual Interpretation of Satellite Imageries
5. Image Processing:Polynomial Rectification-Georeferencing
6. Layer Stacking and Resolution Merge
7. Preparing Image Subset and Mosaic
8. Digital Image Interpretation; Pixel and Histogram Analysis
9. Supervised Classification
10. Unsupervised Classification

LIST OF READINGS

Essential Readings:

1. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
2. Joseph, G. *Fundamentals of Remote Sensing*, 2nd Edition, Universities Press, Hyderabad, 2005.
3. Lillesand, Kiefer and Chipman, *Remote Sensing and Image Interpretation*, 7th Ed., Wiley, 2015.
4. Liu, Jian Guo, and Mason, Philippa J. *Image Processing and GIS for Remote Sensing: Techniques and Applications.* Germany, Wiley, 2016.
5. Nelson, A.C. and Siamak, K. *Image Processing and Data Analysis with ERDAS IMAGINE.* United States, CRC Press, 2018.

Suggested Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.
2. Rees, W.G., *Physical Principles of Remote Sensing,* 3rd Ed., Cambridge University Press, 2012.
3. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer Verlag Berlin Heidelberg, 2013.
4. Schowengerdt, R.A. *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press, San Diego, 2017.

GI 106
GIS SOFTWARE AND DATABASE MANAGEMENT

Course Code: GI 106	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Learn working with GIS software and geodatabase management
- Apply GIS tool and techniques for spatial data collection and editing
- Analyse spatial data and extract spatial information using geospatial techniques

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Introduction to GIS Software
2. Georeferencing Maps/Images
3. Digitization and Topology Creation: Point, Line and Polygon Features
4. Attribute Tables: Creation, Editing, Joining, statistical measurements
5. Query Building: Spatial and Attribute Query
6. Proximity Analysis and Techniques
7. Overlay Operations: Raster and Vector Overlay, Map Algebra
8. Basic Geoprocessing and Data Management
9. Symbolizing and Map Layouts

LIST OF READINGS

Essential Readings:

1. Longley, P. A., Goodchild, M. F., Maguire, D. J., Rhind, D. W. *Geographic Information Science and Systems*. Germany, Wiley, 2015.
2. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.
3. Lo, C.P., and Yeung, Albert K.W. *Concepts and Techniques of Geographic Information Systems*. United Kingdom, Pearson Prentice Hall, 2007.
4. Farkas, Gabor. *Practical GIS.* United Kingdom, Packt Publishing, 2017.
5. Franz Pucha-Cofrep, Andreas F., Fulgencio Canovas-Garcia, F.O. Valdivieso, V. Gonzalez-Jaramillo, and D. Pucha-Cofrep. *Fundamentals of GIS: Applications with ArcGIS*. 2018.

Suggested Readings:

1. Brewer, C.A., *Designing Better Maps: A Guide for GIS Users*, ESRI Press, 2005.
2. Burrough, P. A., McDonnell, R. A. and Lloyd, C.D., *Principles of Geographical Information Systems*, 3rd. ed., Oxford University Press, Toronto, 2016.
3. ESRI: *Esri Map Book*, Volume 32, Esri Press, 2017.
4. Pinde Fu, P.*Getting to Know Web GIS*, Esri Press, 2018.
5. Wise, S., *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.

GI 107
GNSS AND TOTAL STATION SURVEY (Practical/Lab Work)

Course Code: GI 107	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Comprehend basic concepts and theory of GNSS
- Setup GNSS and total station system to conduct field survey
- Collect and analyse field data using GNSS and total station

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Familiarization with GNSS equipments and software.
2. Setting up GNSS receivers and conducting static and kinematic survey
3. GPS based measurement and mapping of terrain features (point, line polygon)
4. DGPS based measurement and mapping of terrain features (point, line polygon)
5. Post-processing GNSS data for accuracy improvement.
6. Total Station- Features, Set-up and calibration.
7. Determination of height, elevation, distance using total station
8. Drawing Contours Using Total Station
9. Measuring and Mapping terrain features (point, line polygon) with Total Station.
10. Performing stakeout operations

LIST OF READINGS

Essential Readings:

1. Christopher J. Hegarty, Elliott D. Kaplan (Eds.). *Understanding GPS/GNSS: Principles and Applications,* Third Edition. United Kingdom: Artech House Publishers.2017.
2. El-Rabbani, A. *Introduction to GPS: The Global Positioning System*, Artech House, 2002.
3. Gopi, S. *Advanced Surveying: Total Station, GIS and Remote Sensing.* India: Pearson Edu. 2007.
4. Leick, A., Rapoport, L., Tatarnikov, D. *GPS Satellite Surveying*. Germany: Wiley. 2015
5. Letham, L., and Letham, A. *GPS Made Easy: Using Global Positioning Systems in the Outdoors.* Australia: Rocky Mountain Books. 2011.

Suggested Readings:

1. Kavanagh, B. F.&Mastin, T. B. *Surveying: Principles and Applications*. UK: Pearson. 2014
2. Noureldin. A., Karamat, T.B. and Jacques Georgy, *Fundamentals of Inertial Navigation, Satellite-based Positioning and their Integration*, Springer, 2013.
3. Rao, G.S., *Global Navigation Satellite Systems- With Essentials of Satellite Communications*, Tata McGraw Hill Education Private Limited, New Delhi, 2010.
4. Schofield, W. and Breach, M. *Engineering Surveying*. United Kingdom: CRC Press. 2007
5. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

GI 108
OPEN-SOURCE GIS

Course Code: GI 108	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Understand the concept and usage of open-source GIS
- Develop ability to create, analyse and interpret geospatial data
- Reproduce and visualize mapsusing spatial data in open-source GIS

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Familiarisation with QGIS
2. Importing spatial data formats into QGIS
3. Digitizing point, line and polygon
4. Creating raster dataset
5. Crating spatial database
6. Vector data processing and analysis
7. Raster data processing and analysis
8. Using plugins for enhanced data visualization
9. Creating thematic maps and customizing map layouts
10. Exporting maps and data from QGIS to different formats

LIST OF READINGS

Essential Readings

1. Cutts, A. and Graser, A. *Learn QGIS: Your Step-by-step Guide to the Fundamental of QGIS 3.4*, 4th Edition. Germany, Packt Publishing, 2018.
2. Cutts, A. *QGIS Quick Start Guide: A Beginner's Guide to Getting Started with QGIS3.4*. Germany, Packt Publishing, 2019.
3. Madry, S. *Introduction to QGIS: Open-Source Geographic Information System*. United States. Locate Press. 2021
4. Graser, A., and Peterson, G. N. *QGIS Map Design: With New and Updated Workflows for QGIS 3.4*. United States: Locate Press. 2018.

Suggested Readings

1. Bruy, A., and Svidzinska, D. *QGIS By Example*. United Kingdom: Packt Publishing. 2015.
2. Lawhead, Joel. *QGIS Python Programming Cookbook*. India, Packt Publishing, 2017.
3. Mearns, B. *QGIS Blueprints*. United Kingdom: Packt Publishing. 2015
4. Menke, K., Smith, R., Jr., R. S., Pirelli, L., Van Hoesen, J. *Mastering QGIS: Go Beyond the Basics and Unleash the Full Power of QGIS with Practical, Step-by-step Examples*. India: Packt Publishing. 2016.

GI 109
JAVASCRIPT AND SQL

Course Code: GI 109	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Comprehend fundamental programming concepts and methodology
- Learn and apply Java Script and SQL programming
- Read, understand and trace the execution of programs written in Java and SQL

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Introduction to JavaScript syntax and programming concepts.
2. Writing basic scripts for web mapping applications using Leaflet or OpenLayers.
3. Interactive maps with basic functionality like markers and popups using JavaScript libraries.
4. Implement basic layers (vector and raster layers) on maps to visualize spatial data.
5. Introduction to SQL syntax and database concepts.
6. Writing basic SQL queries for data retrieval and manipulation.
7. Use SQL queries to filter and sort spatial and non-spatial data based on different conditions
8. Write SQL queries to join multiple tables and aggregate data

LIST OF READINGS

Essential Readings

1. Aitchison, A. *Beginning Spatial with SQL Server 2008.* United States: Apress. 2009
2. Fehily, C. *SQL: Visual QuickStart Guide.* United Kingdom: Pearson Education. 2010.
3. Obe, R., Hsu, L. S. *PostGIS in Action,* Third Edition. United Kingdom: Manning. 2021
4. Online Resources: www.w3schools.com for HTML and SQL. (*Live and latest).*
5. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
6. Zandbergen, P. A. *Python Scripting for ArcGIS.* United States: ESRI Press. 2013.

Suggested Readings

1. Lovelace, R., Nowosad, J., Muenchow, J. *Geocomputation with R.* US: CRC Press. 2019.
2. Lutz, M. *Programming Python*, 3rd Edition, O'Reilly Media, Inc. 2006.
3. Tateosian, L. *Python for ArcGIS*, Springer, 2015
4. Yang, C. *Introduction to GIS Programming and Fundamentals with Python and ArcGIS*, CRC Press, 2017.
5. Bivand, Roger S., Pebesma, Edzer, and Gomez-Rubio, V. *Applied Spatial Data Analysis with R.* Netherlands: Springer New York. 2013
6. Brunsdon, C., Comber, L. *An Introduction to R for Spatial Analysis and Mapping*. United Kingdom: SAGE Publications. 2014

GI 110
STATISITICAL TECHNIQUES AND ANALYSIS

Course Code: GI 110	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Develop analytical thinking on statistical measures and interpretation of geospatial data
- Visualize spatial data distributions through histograms and frequency tables.
- Analyze probability distributions, hypothesis testing, data driven informed decisions.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Data: Spatial & Non-spatial, Grouped &Ungrouped, Continuous &Discrete
2. Descriptive Statistics
   a. Measures of Central Tendency: Mean, Median and Mode
   b. Measures of Dispersion: Range, variance, standard deviation, coefficient of variability
3. Normal Distribution: Characteristics, measurements and applications; z-distribution
4. Inferential Statistics:
   a. Parametric statistical test: T-Test, Z-Test and ANOVA
   b. Non-parametric statistical test: Chi-Square Test
   c. Correlation: Scatterplot, Spearman's method and Pearson's correlation
   d. Regression analysis, construction of regression line.

LIST OF READINGS

Essential Readings:

1. Burt, J.E., G.M. Barber and D.L. Rigby.*Elementary Statistics for Geographers*, Guilford Press, New York, 2009.
2. Clark, W.A.V. and P.L. Hosking.*Statistical Methods for Geographers*, Wiley, New York, 1986.
3. Gupta, S. P.: *Statistical Methods*, 46th Edition, Sultan Chand and Sons, New Delhi, 2021.
4. Peter, J. Taylor.*Quantitative Methods in Geography*,Houghton Mifflin Company, Boston, 1977.
5. Hammond and McCullagh.*Quantitative Methods in Geography*, Clarendon Press, Oxford, 1974.

Suggested Readings:

1. David M. Smith.*Patterns in Human Geography*, Penguin, Harmondsworth, 1975.
2. Montello, D.R. and Sutton, P.C. *An Introduction to Scientific Research Methods in Geography*, Sage Publication Inc., California, 2006.
3. Ott, R. Lyman and Michael Longnecker.*An Introduction to Statistical Methods and Data Analysis*, Brooks/Cole, USA, 2010.
4. Rogerson, P. A. *Statistical Methods for Geography: A Student's Guide*, Sage Pub., London, 2015.
5. Sarkar, A.*Quantitative Geography: Techniques and Presentations*, Orient Blackswan, US, 2013.

SEMESTER-II

GI 201
INTRODUCTION TO CARTOGRAPHY

Course Code: GEOG 201	***Maximum Marks: 50***
Course Type: Core	Terminal Examination: 40 Marks
Course Credit: 2	Internal Assessment: 10 Marks
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Trace and contextualize the evolution of cartographic tradition
- Understand theoretical principles of cartography
- Develop skills for map designs using cartographic principles
- Apply cartographic knowledge for representation and interpretation of information.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit I

i. Cartography: Nature, scope and historical evolution
ii. Computer Assisted Cartography

Unit II

iii. Types of Data and Data Sources
iv. Elements of Map Reading: Interpreting Physical and Cultural Features

Unit -III

v. Elementary conventional methods: Contours, Spot heights, Hachures, Hill shading
vi. Profiles: Transverse and Longitudinal

Unit -IV

vii. Representation of Population data and Agricultural data
viii. Representation of Climatic data: Temperature and Rainfall

LIST OF READINGS

Essential Readings

1. Bolstad, P. *GIS Fundamentals: A First Text on Geographic Information Systems*, Second Edition, White Bear Lake, MN: Eider Press, 2005.
2. Dodge M.*Cartographic Principles: Map Design*,Routledge, 2015
3. Field, K.*Cartography*, Esri Press, 2018
4. Harvey, F. *A Primer of GIS, Fundamental Geographic and Cartographic Concepts*, The Guilford Press, 2008.
5. Robinson, A. H., Morrison, J. L., Phillip C. Muehrcke, A. Jon Kimerling, Stephen C. Guptill. *Elements of Cartography*, 6th Edition, Wiley India Pvt. Limited, 2009.

Suggested Readings

1. Crampton, W. C. *Mapping: A Critical Introduction to Cartography and GIS*, John Willy &Sons, New York, 2010.
2. Keates, J.S. *Cartographic Design and Production*, Longman, London, 1998.
3. Maltiar K.K and Maltiar S.R., *Concepts of Cartography, Remote Sensing and GIS*, Rajesh Publications, 2019
4. Krygier,J. and Wood, D. *Making Maps: A Visual Guide to Map Design for GIS*, Guilford Publications, 2013.

GI 202
SPATIAL STATISTICS AND ANALYSIS

Course Code: GI 202	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Define and explain fundamental conceptsof spatial analysis and its applications
- Evaluate and summarise spatial analysis tools and techniques
- Apply spatial analysis techniques for extraction and interpretation of information

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit- I

Fundamentals of Spatial Analysis

i. Concept and Significance of Spatial Analysis
ii. Point Pattern Analysis
   - Spatial central tendency and dispersion
   - Spatial Variations in point intensity; Measures based on distance between events

Unit- II

Spatial Techniques for Line Feature Analysis

iii. Measurement of Linear Features over Space
iv. Network Analysis
   - Point-to-point Analysis; Coverage analysis
   - Location-allocation

Unit-III

Spatial Statistical Techniques

v. Hot-spot analysis, Spatial Autocorrelation
vi. Nearest Neighbour Analysis
vii. Spatial Regression and Correlation

Unit-IV

Raster Analysis: Elevation Models and Surface Analysis

viii. Elevation Models: DEM, DTM, DSM and TIN
ix. Surface Analysis: Characterisitcs, Types and Representation
x. Surface Creation: Interpolation Techniques

LIST OF READINGS

Essential Readings:

1. Chang, Kang-tsung, *Introduction to Geographic Information Systems,* 8th Edition,Tata McGraw-Hill Education, 2015.
2. Fotheringham Stewart A. and Rogerson A. Peter, *The SAGE Handbook of Spatial Analysis*, SAGE Publications, London, 2009.
3. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, London: Academic Press, 1994.
4. Sanders, L., *Models in Spatial Analysis*, John Wiley & Sons, 2013.
5. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.

Further Readings:

1. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
2. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2ndEdition, John Wiley and Sons, 2000.
3. Lloyd, Christopher D., *Spatial Data Analysis: An Introduction for GIS Users*, Oxford University Press Inc., New York, 2010.
4. Maguire, Batty and Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
5. Murayama, Y. and R. Thapa, *Spatial Analysis and Modeling in Geographical Transformation Process: GIS Based Applications*, Springer, 2011.

GI 203
ADVANCES IN DIGITAL IMAGE PROCESSING

Course Code: GI 203	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Comprehend techniques of image processing and analysis
- Apply various techniques of image restoration, enhancement and classification
- Compute various indices and extract information using satellite imageries

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S.No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit- I

Digital Image Pre-processing

i. Concept and Characteristics of Digital Image and Image Processing
ii. Radiometric Correction: Errors, Sources, Types and Correction Methods
iii. Geometric Correction: Errors, Sources, Types and Correction Methods.

Unit-II

Image Enhancement Techniques

iv. Introduction to Image Enhancement Techniques
v. Radiometric Enhancement Techniques
vi. Spatial Enhancement: Filtering Techniques

Unit-III

Image Transformation: Spectral Enhancement

vii. Image Arithmetic and Spectral Indices
viii. Principal Component Analysis (PCA)
ix. Intensity, Hue and Saturation; Fourier Transformation

Unit- IV

Digital Image Classification

x. Image Classification Approaches and Techniques
xi. Information Extraction using Artificial Intelligence
xii. Accuracy Assessment; Change Detection and Analysis

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing*, Guilford Press, 2012.
2. Chen, C.H., *Image Processing for Remote Sensing,* CRC Press, 2008.
3. Gonzalez, R.C. & Woods, R.E., *Digital Image Processing*, 2nd edition, Prentice-Hall, Inc., 2002.
4. Liu, J.G. and Mason, P.J., *Essential Image Processing and GIS for Remote Sensing,* John Wiley & Sons, 2009.
5. Mather, P.M. & Magaly, K., *Computer Processing of Remotely-Sensed Images: An Introduction,* Fourth Edition, John Wiley & Sons, 2011.

Suggested Readings:

1. Dong, P. and Qi Chen, *LiDAR Remote Sensing and Applications*. CRC Press, Taylor & Francis Group, 2018.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
3. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
4. Lillesand, T.M., Kiefer, R.W. & Chipman, J. W., *Remote Sensing and Image Interpretation*, Fourth Edition, John Wiley and Sons, 2004.
5. Schowengerdt, Robert A. *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press: San Diego, 2017.

GI 204
FUNDAMENTALS OF PHOTOGRAMMETRY

Course Code: GI 204	***Maximum Marks: 50***
Course Type: Core Course	Theory Paper: 40
Course Credit: 2	Internal Assessment: 10
Teaching Hours/Week: 2	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Understand theory, concepts and working principles of photogrammetry
- Analyse and measure data using photogrammetric techniques
- Apply various techniques of image analysis and information extraction

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any six short questions in about 25-30 words each. Each part shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at S. No. 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment.

COURSE CONTENT

Unit - I

i. Fundamentals of Photogrammetry: Definition and Types
ii. Historical Developments of Photogrammetry; Application of Photogrammetry

Unit - II

iii. Aerial Photographs: Types and Geometry; Elements of Photo Interpretation
iv. Principles of Stereoscopy; Stereoscopic Viewing and Parallax; Orthophoto rectification

Unit - III

v. Elementary Methods of Planimetric Mapping
vi. Topographic Mapping and Spatial Data Collection

Unit - IV

vii. Advances in Photogrammetry; Drone Technology and Integration with LiDAR
viii. Flight planning parameters and site selection

LIST OF READINGS

Essential Readings:

1. Linder, W. *Digital photogrammetry a practical course*. Berlin: Physica-Verlag. 2006
2. McGlone, J. C., *Manual of Photogrammetry,* American Society of Photogrammetry and Remote Sensing, 2013.
3. Mikhail, Bethel, and McGlone.*Introduction to Modern Photogrammetry*, India: Wiley. 2001.
4. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

Suggested Reading:

1. Burnside, C. D., *Mapping from Aerial Photography,* 2nd Ed, Collins, 1985.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.
3. Sabins, F. F. *Remote Sensing: Principles and Interpretation,* New York: 2007.
4. Wolf, P. R., Wolf, P., DeWitt, B. A., Wilkinson, B. E. *Elements of Photogrammetry with Application in GIS*, Fourth Edition. United States: McGraw. 2013

GEOG 205
CARTOGRAPHIC TECHNIQUES AND VISUALISATION

Course Code: GI205	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Demonstrate practical understanding of mapping techniques
- Design map outputs using cartographic principles
- Interpret mapped information,and relate maps to the real world in spatial context

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab Work

1. Classification of Topographical Sheets
2. Methods of landform analysis: Contours, Spot heights, Hachures, Hill shading
3. Profiles and cross-sections: Transverse and Longitudinal
4. Methods of slope analysis and interpretation (Wentworth and Robinson)
5. Mapping Techniques and Design: Dot Map, Choropleth Map, Isopleth Map
6. Constructing statistical diagrams: Bar Graphs, Line Graphs, Pie Charts

LIST OF READINGS

Essential Readings

1. Dodge M.*Cartographic Principles: Map Design*,Routledge, 2015
2. Field, K.*Cartography*, Esri Press, 2018
3. Harvey, F. *A Primer of GIS, Fundamental Geographic and Cartographic Concepts*, The Guilford Press, 2008.
4. Robinson, A. H., Morrison, J. L., Phillip C. Muehrcke, A. Jon Kimerling, Stephen C. Guptill. *Elements of Cartography*, 6th Edition, Wiley India Pvt. Limited, 2009.

Suggested Readings

1. Crampton, W. C. *Mapping: A Critical Introduction to Cartography and GIS*, John Willy &Sons, New York, 2010.
2. Maltiar K.K and Maltiar S.R., *Concepts of Cartography, Remote Sensing and GIS*, Rajesh Publications, 2019
3. Singh, R. L. and Singh, R.P.B.*Elements of Practical Geography*, Kalyani Pub., Delhi, 2005.
4. Krygier,J. and Wood, D. *Making Maps: A Visual Guide to Map Design for GIS*, Guilford Publications, 2013.

GI 206
SPATIAL ANALYSIS TOOLS AND TECHNIQUES

Course Code: GI 206	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Define and explain fundamental conceptsof spatial analysis
- Evaluate and summarise spatial analysis tools and techniques
- Apply spatial analysis for extraction and interpretation of information

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.

2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab Work

1. Point Pattern Analysis:
   - Measuring Mean Centre, Weighted Mean Centre and Median Centre
   - Measuring Standard Distance and Standard Distance Ellipse
   - Nearest Neighbour Analysis
2. Network Analysis
   - Point-to-point Analysis: nearest location, shortest distance, fastest route
   - Coverage: determining service area
   - Optimal Site Selection: Location-allocation, Site Suitability, Route Optimization
3. Measuring Spatial Autocorrelation
4. Hot-spot Analysis
5. Spatial Interpolation Techniques and Analysis
6. DEM Creation using Point data, Contours and Lidar Data
7. Spatial Simulation: Model Building in GIS
8. Spatial Multi-Criteria Decision Analysis (MCDA)

LIST OF READINGS

Essential Readings:

1. Chang, K.*Introduction to Geographic Information Systems,* Tata McGraw-Hill Edu., 2015.
2. Fotheringham and Rogerson.*The SAGE Handbook of Spatial Analysis*, SAGE, London, 2009.
3. Laurini and Thomson, *Fundamentals of Spatial Information Systems*, Academic Press, 1994.
4. Sanders, L., *Models in Spatial Analysis*, John Wiley & Sons, 2013.
5. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.

Suggested Readings:

1. Burrough and Macdoneli. *Principles of Geographic Information Systems,* Oxford University Press, 2000.
2. Demers, M.N., *Fundamentals of Geographic Information Systems*, John Wiley and Sons, 2000.
3. Lloyd, Christopher D., *Spatial Data Analysis: An Introduction for GIS Users*, Oxford University Press Inc., New York, 2010.
4. Maguire, Batty and Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.

GI 207
DIGITAL IMAGE PROCESSING TECHNIQUES

Course Code: GI 207	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Define and explain fundamental conceptsof digital image processing
- Perform image rectification and radiometric correction techniques
- Apply image enhancement and classification techniques

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab Work

1. Geometric and Radiometric Corrections
2. Image Enhancement and Spatial Filtering Techniques
3. Spectral Indices: Vegetation, Water and Landscape Indices
4. Image Transformation Techniques
5. Image Classification, Accuracy Assessment and Change Detection
6. Object-Oriented Modeling (OOM)
7. Hyperspectral Data: Acquisition, Preprocessing, Analysis and Feature Extraction
8. SAR Data: Acquisition, Preprocessing, Image Classification and Feature Extraction

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne.*Introduction to Remote Sensing*, Guilford Press, 2012.
2. Chen, C.H. *Image Processing for Remote Sensing,* CRC Press, 2008.
3. Gonzalez, R.C. & Woods, R.E. *Digital Image Processing*, 2nd edition, Prentice-Hall, Inc., 2002.
4. Liu, J.G. and Mason, P.J. *Essential Image Processing and GIS for Remote Sensing,* John Wiley & Sons, 2009.
5. Mather, P.M. & Magaly, K.*Computer Processing of Remotely-Sensed Images: An Introduction,* Fourth Edition, John Wiley & Sons, 2011.

Suggested Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
2. Richards, J.A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer Verlag Berlin Heidelberg, 2013.
3. Schowengerdt, Robert A. *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press: San Diego, 2017.

GI 208
PHOTOGRAMMETRY AND DRONE TECHNOLOGY

Course Code: GI 208	***Maximum Marks: 50***
Course Type:Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Outline the use of photogrammetry and drone technology
- Apply photogrammetric techniques for information collection
- Undertake surveys and data collection using drone technology

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.

2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab Work

1. Simple Geometry of Aerial Photographs
2. Information on the Margins of Aerial Photograph
3. Stereoscope and its orientation; Stereo-vision test; 3D Viewing
4. Locating principal point, conjugate principal point, flight line
5. Stereoscopic parallax; Relief Displacement
6. Measurement of Scale, Height and Slope
7. Identification and mapping of features.
8. Working with Drone: Mission Planning, Data Capturing
9. Using Drone Software: Data Processing and Mapping

LIST OF READINGS

Essential Readings:

1. Linder, W. *Digital Photogrammetry: Theory and Applications*. Germany: Springer Berlin Heidelberg. 2013
2. Linder, W. *Digital photogrammetry a practical course*. Berlin: Physica-Verlag. 2006
3. McGlone, J. C., *Manual of Photogrammetry (6th Edition),* American Society of Photogrammetry and Remote Sensing, 2013.
4. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014

Suggested Reading:

1. Burnside, C. D., *Mapping from Aerial Photography,* 2nd Ed, Collins, 1985.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.
3. Joseph, George, *Fundamentals of Remote Sensing,* Universities Press India, 2007.
4. Wolf, P. R., Wolf, P., DeWitt, B. A., Wilkinson, B. E. *Elements of Photogrammetry with Application in GIS*, Fourth Edition. United States: McGraw. 2013

GI 209
PROGRAMMING IN PYTHON AND R

Course Code: GI 209	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Outline the application of python and R language in geoinformatics
- Apply python on spatial data analysis and interpretation
- Undertake data analysis using R language and software environment

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

Python Exercises

1. Introduction to Python Programming
2. Control Structures and Loops
3. Functions and Modules
4. File Handling
5. Introduction to Geospatial Libraries
6. Plotting Geospatial Data

R Exercises

7. Introduction to R Programming
8. Control Structures and Loops
9. Functions and Packages
10. Data Import and Export
11. Introduction to Geospatial Libraries
12. Plotting Geospatial Data

LIST OF READINGS

Essential Readings

1. Brunsdon, C., Comber, L. *An Introduction to R for Spatial Analysis and Mapping*. United Kingdom: SAGE Publications. 2014
2. Lovelace, R., Nowosad, J., Muenchow, J. *Geocomputation with R.* US: CRC Press. 2019.
3. Lutz, M. *Programming Python*, 3rd Edition, O'Reilly Media, Inc. 2006.
4. Tateosian, L. *Python for ArcGIS*, Springer, 2015

Suggested Readings

1. Bivand, R.S., Pebesma, E., and Gomez-Rubio, V. *Applied Spatial Data Analysis with R*. Netherlands: Springer New York. 2013.
2. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
3. Yang, C. *Introduction to GIS Programming and Fundamentals with Python and ArcGIS*, CRC Press, 2017.
4. Zandbergen, P. A. *Python Scripting for ArcGIS.* United States: ESRI Press. 2013.

GI 210
WEB GIS

Course Code: GI 210	***Maximum Marks: 50***
Course Type: Core Course (Practical)	Practical Record: 20 Marks
Course Credit: 2	Viva-Voce: 20 Marks
Teaching Hours/Week: 4	Internal Assessment: 10 Marks
	Examination Time: 3 Hours

COURSE OUTCOMES: The course would help students to:

- Develop understanding of basic principles and techniques of Web GIS
- Hands on Training using Web GIS technologies to execute Web GIS projects
- Design web architecture using geospatial database, mapping servers, and libraries

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. WebGIS-Basic Concept and Advantages
2. Open-source Libraries for Mapping and Analysis
3. Setting up a Basic Web-map
4. Add Map Objects
5. Adding Popups
6. Adding and Managing Controls
7. Raster Data: Adding Basemaps and Overlays
8. Vector Data: Adding Marker, Path/line, Polygon; Draw custom shapes
9. Styling Vector Data
10. Searching, filtering and analysing data
11. Basic Spatial Analysis
12. Legend Creation
13. Publishing a web map

LIST OF READINGS

Essential Readings

1. DuVander, A. *Map Scripting 101: An Example-driven Guide to Building Interactive Maps with Bing, Yahoo!, and Google Maps*. United States: No Starch Press. 2010.
2. Fu, P. *Getting to Know Web GIS*. United States: Esri Press. 2022.
3. Fu, P. and Jiulin Sun. *Web GIS Principles and Applications*. Redlands: Esri Press, 2011. Kraak, M. and A. Brown. *Web Cartography: Developments and Prospects*. London: Taylor and Francis, 2001.
4. Ngugi, S. G.*Leaflet in Practice: Create webmaps using the JavaScript Leaflet library*. R Bookdown. 2023.

Suggested Readings

1. Davis, S. *GIS for Web Developers: Adding where to Your Web Applications*. United States: Pragmatic Bookshelf. 2007.
2. Fu, P. *Getting to Know Web GIS, Third Edition*. Redlands: Esri Press, 2018.
3. Shellito, B. *Introduction to Geospatial Technologies, 4th Ed.* New York: W. H. Freeman, 2017.
4. Stefanakis, E. *An introduction to web mapping and geospatial web services*. 2015. Available at the Geolibrary.